

# Pratik J. Gandhi's CV

pratik.gandhi@yale.edu | (929) 529-0467 | [Personal website](#)  
Postdoctoral Associate, Department of Astronomy, Yale University

## RESEARCH INTERESTS

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theoretical astrophysics and galaxy formation  
cosmological zoom-in simulations of galaxies  
near-field cosmology and connections to cosmic reionisation  
low-mass galaxies across cosmic time  
galactic archeology and stellar fossil record of Local Group galaxies  
synthetic observations and catalogs from galaxy simulations  
pop III star formation, feedback, and enrichment

## APPOINTMENTS

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|-------------------------------------------------------------------|-------------------|
| Postdoctoral Associate – Department of Astronomy, Yale University | Nov '24 – Present |
| Graduate Student – Department of Physics & Astronomy, UC Davis    | Sep '18 – Oct '24 |

## EDUCATION

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|-------------------------------------------------------------------|--------------------|
| Ph.D. in Physics (Astrophysics) – University of California, Davis | Sep '18 – Oct '24  |
| M.S. in Physics (Astrophysics) – University of California, Davis  | Sep '18 – June '20 |
| B.S. in Astrophysics – Yale University                            | Aug '14 - May '18  |

## FELLOWSHIPS & AWARDS

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|-------------------------------------------------------------|-----------------|
| Outstanding Graduate Student Teaching Award   UC Davis      | May'23          |
| FAMOUS travel grant   AAS                                   | Dec'22          |
| Frontera Computational Science Fellowship   TACC            | Jun'22 - May'23 |
| APS WiP Grant   for UCD Diversity & Inclusion in Phys/Astro | Jan'21 - Jan'23 |
| Graduate Program Research Fellowship   UC Davis             | Jan'21 - Apr'21 |
| Paul Brady & Charles Soderquist Fellowship   UC Davis       | Sep'18 - Aug'19 |
| Pierson College Mellon Forum Award   Yale U.                | Aug'17 - May'18 |
| Tetelman Fellowship for International Research   Yale U.    | Summer 2016     |
| Pierson College Richter Fellowship   Yale U.                | Summer 2016     |
| Freshman Summer Research Fellowship   Yale U.               | Summer 2015     |

## PUBLICATIONS, FIRST AUTHOR OR LEADING CONTRIBUTION

([ADS Library](#))

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[4] **Gandhi, P. J.**, Wetzel, A., Boylan-Kolchin, M., Sanderson, R. E., et al., “*Testing the near-far connection with FIRE simulations: UVLFs, comparison to high-redshift simulations, and distance dependence*” (submitted to ApJ)

- [3] Wetzel, A., Samuel, J., **Gandhi, P. J.**, et al., *Second public data release of the FIRE-2 cosmological zoom-in simulations of galaxy formation*, arxiv:2508.06608, 2025
- [2] **Gandhi, P. J.**, Wetzel, A., Boylan-Kolchin, M., Sanderson, R. E., et al., “*Testing the near-far connection with FIRE simulations: inferring the stellar mass function of the proto-Local Group at  $z > 6$  using the fossil record of present-day galaxies*”, MNRAS, 533, 1059G, 2024
- [1] **Gandhi, P. J.**, Wetzel, A., Hopkins, P. F., Shappee, B. J., et al., “*Exploring metallicity-dependent rates of Type Ia supernovae and their impact on galaxy formation*”, MNRAS, 516, 1941G, 2022

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## PUBLICATIONS, IN PREP

- [1] **Gandhi, P. J.**, Bellinger, E., Geha, M., Wetzel, A., Hopkins, P. F., et al., “*Modelling stellar feedback from the first stars in the FIRE-3 cosmological zoom-in simulations*”, (in prep for submission to ApJ)

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## PUBLICATIONS, CONTRIBUTING AUTHOR

- [6] Kundu, A., Sanderson, R., Lidz, A., **Gandhi, P. J.**, et al., *Rise of the forsaken relics: connecting present-day stellar streams and phase-mixed galaxies to the Epoch of Reionization*, arXiv:2510.17968, 2025 (submitted to ApJ)
- [5] Wheeler, C., Moreno, J., Rodriguez Wimberly, M. K., et al. (including **Gandhi, P. J.**), *How invisible stellar halos bias our understanding of ultra-faint galaxies*, arXiv:2506.15785, 2025 (submitted to ApJ)
- [4] de los Reyes, M. A., Asali, Y., Wechsler, R., (including **Gandhi, P. J.**), *Stellar Mass Calibrations for Local Low-mass Galaxies*, ApJ, 989, 91D, 2025
- [3] Horta, D., Cunningham, E. C., Sanderson, R., et al. (including **Gandhi, P. J.**), *The proto-galaxy of Milky Way-mass haloes in the FIRE simulations*, MNRAS, 527, 9810H, 2024
- [2] Lemaux, B. C., Fuller, S., Bradač, M., Pentericci, L., Hoag, A., et al. (including **Gandhi, P. J.**), “*The size and pervasiveness of ly-UV spatial offsets in star-forming galaxies at  $z \sim 6$* ”, MNRAS, 504, 3662L, 2021
- [1] Pelliccia, D., Strait, V., Lemaux, B. C., Bradač, M., Coe, D., et al. (including **Gandhi, P. J.**), “*RELICS-DP7: Spectroscopic confirmation of a dichromatic primeval galaxy at  $z \sim 7$* ”, ApJ, 908L, 30P, 2021

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## SUPERCOMPUTING TIME AWARDED

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| <b>PI:</b> NSF Frontera (Texas Advanced Computing Center) – 50k node-hours | Jun '22 |
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## CONFERENCE TALKS & PRESENTATIONS

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|--------------------------------------------------------------------|---------|
| <i>Invited talk</i> , Gruber Cosmology Prize Symposium, Yale       | Oct '25 |
| <i>Contributed talk</i> , Santa Cruz Galaxy Workshop, UCSC, CA     | Aug '25 |
| <i>Contributed talk</i> , Galactic Frontiers II, Dartmouth College | Jun '25 |

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|-----------------------------------------------------------------------------------|---------|
| <i>Contributed talk</i> , Tinsley Workshop on early Universe galaxies, Yale       | Oct '24 |
| <i>Contributed talk</i> , GalFRESKA conference, UC Riverside, CA                  | Sep '23 |
| <i>Contributed talk</i> , Santa Cruz Galaxy Workshop, CA                          | Aug '23 |
| <i>Poster presentation</i> , First Light Conference at MIT, MA                    | Jun '23 |
| <i>Poster presentation</i> , IAU 377 Symposium in Kuala Lumpur, Malaysia          | Feb '23 |
| <i>Contributed talk</i> , AAS 240 Summer Meeting in Pasadena, CA                  | Jun '22 |
| <i>Contributed talk</i> , SAZERAC SIP: Galaxy formation near and far              | Dec '21 |
| <i>Contributed talk</i> , ISM 2021: Structures, Scales, and Star formation        | May '21 |
| <i>Senior thesis talk</i> , Yale University Mellon Forum                          | Feb '18 |
| <i>Poster presentation</i> , AAS 231 Winter Meeting in National Harbour, USA      | Jan '18 |
| <i>Poster presentation</i> , Ivy League Undergraduate Research Symposium at UPenn | Nov '17 |
| <i>Contributed talk</i> , Yale Astronomy Summer Symposium                         | Sep '16 |

## INVITED COLLOQUIA AND SEMINARS

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|-------------------------------------------------|---------|
| Physics & Astronomy colloquium, Amherst college | Nov '25 |
| Caltech Astronomy tea talk                      | Apr '25 |
| Carnegie Observatories lunch seminar            | Apr '25 |
| Yale Astronomy Galaxy Lunch                     | Apr '24 |
| Stanford KIPAC Tea Talk                         | Sep '23 |
| Pittsburgh-CMU joint astro lunch seminar        | Sep '23 |
| UT Austin Astronomy                             | May '23 |
| UPenn Galaxy Dynamics group                     | Feb '23 |
| UC Davis Physics & Astronomy                    | Oct '22 |
| Carnegie Tea at Carnegie Observatories          | Mar '22 |
| UH Institute for Astronomy (IfA) journal club   | Feb '22 |

## PUBLIC TALKS

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|------------------------------------------------------------|----------|
| <i>Invited</i> , Museum of Science & Curiosity, Sacramento | May '22  |
| <i>Invited</i> , Mt. Diablo Astronomical Society           | July '20 |

## RESEARCH (CO-)ADVISING

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|                                           |                   |
|-------------------------------------------|-------------------|
| Graduate Students (2)                     |                   |
| 1. Bhavya Pardasani (Yale University)     | Jun '25 - Present |
| 2. Yasmeen Asali (Yale University)        | Jan '25 - Present |
| Undergraduate Students (3)                |                   |
| 1. Grace Gogglietino (Colgate University) | Jun '25 - Present |
| 2. Alfredo Calderon (Cal Poly Humboldt)   | Jun '23 - Sep '23 |
| 3. Heather Pearson (Oberlin College)      | Jun '22 - Jun '23 |

## TEACHING

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*Lead Teaching Assistant | UC Davis*

- Astrophysics of Galaxies Winter '22
- Undergraduate Astronomy Lab Spring '19
- Introduction to Astronomy Fall '19

*Teaching Assistant | UC Davis*

- Physics for Life Sciences Fall '18, Winter '19, Summer '19, Winter '20, Spring '20

*Organiser | Sacramento School District Astronomy Olympiad* Jan '19, Jan '20

## SERVICE WORK & OUTREACH

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*Journal review, Open Journal of Astrophysics (OJA)*

*UAW Labor Summer* Jun '23 - Sep '23

*DEI Committee co-chair, Astrobites blog (AAS)* Sep '21 - Aug '22

*Author, Astrobites blog (AAS)* Dec '20 - Sep '24

*Organizer, Astronomy on Tap, Davis (science cafe)* Jan '21 - Sep '24

*Chair, UCD Diversity & Inclusion in Physics & Astronomy* Sep '21 - Sep '24

*Graduate student rep, UCD Physics/Astronomy Graduate Admissions* Sep '21 - Jan '22

*Graduate student rep, UCD Physics/Astronomy Climate Survey* Aug '20 - Apr '21

*Member, UCD Diversity & Inclusion in Physics & Astronomy* Sep '19 - Aug '21

## PROGRAMMING

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**Languages/Platforms:** python, C, Java, Unix/Linux/MacOS, Windows, slurm

**Packages:** numpy, scipy, astropy, matplotlib, MPI, OpenMP, GIZMO, Lenstool, SWUnited